

DPEN022: Enabling Mathematics B

Class Questions Week 5

Week 5A

Key Idea – Definite Integrals of Power of x

1. Evaluate the following definite integrals:

(a) $\int_{-1}^2 x^3 dx$

(b) $\int_{-1}^1 x(1-x^2) dx$

(c) $\int_{-1}^0 2x^5 + x - 7 dx$

(d) $\int_2^4 \left(x + \frac{1}{x}\right)^2 dx$

(e) $\int_3^4 \frac{5x^4(x+2)}{x^3} dx$

(f) $\int_{-5}^5 (x-1)(x^2+x+1) dx$

(g) $\int_0^4 x^{\frac{1}{2}} dx$

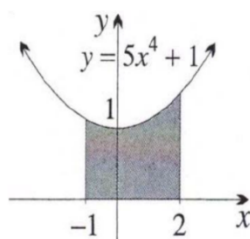
(h) $\int_1^{27} \frac{1}{2\sqrt[3]{x^2}} dx$

(i) $\int_0^1 (\sqrt{x} - \sqrt[3]{x}) dx$

Key Idea – Area under the Curve

For each graph below, find the area of the shaded region.

1. (a)

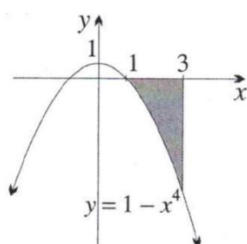


2. (a) Find the area of the region bounded between the curve $y = x + 1$ and the x -axis from $x = -1$ to $x = 1$.
- (b) Find the area of the region bounded between the curve $y = 1 - x^2$ and the x -axis from $x = -0.5$ to $x = 0.5$.
- (c) Calculate the area of the region bounded by the curve $y = x^2 - 4x + 4$ and the x -axis and the lines $x = 1$ and $x = 4$.

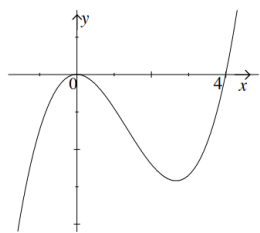
Week 5B

Key Idea – Area below the x -axis and Subdividing Regions before Integrating

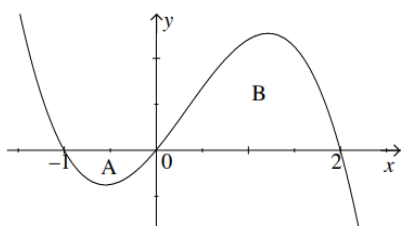
1. For the graph below, find the area of the shaded region.



2. Find the area enclosed by the graph of $y = 3x^2(x - 4)$ and the x -axis.

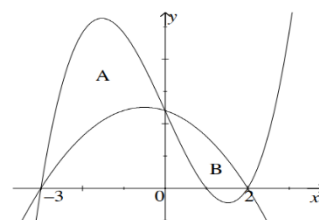


3. Given the function $y = x^2 - 6$.
- Find the x -intercepts of the curve.
 - Sketch the graph.
 - Find the area of the region between the curve and the x -axis where the curve is below the x -axis.
4. Find the area of the region bounded between the curve $y = x^2 - 6$ and the x -axis from $x = 1$ to $x = 3$.
5. Find the total area enclosed between the graph of $y = -12x^3 + 12x^2 + 24x$ and the x -axis.

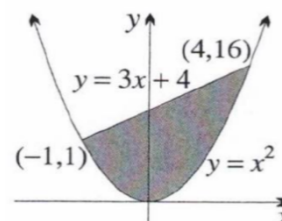
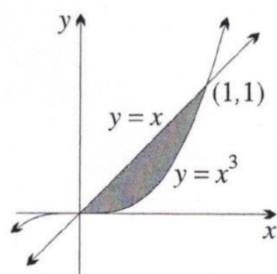


Key Idea – The Area between two Curve.

1. Given the graphs of the function $y = 6 - x - x^2$ and $y = x^3 - 7x + 6$ (upper curve).
- Find the points of intersection of the curves.
 - Find the total area enclosed between them.



2. (a) (b)



3. Find the area enclosed by the curves $y = x^2$ and $y = -x^2 + 4x$.
4. Find the exact area enclosed between the parabola $y = x(x - 2)$ and the line $y = -x + 2$.
5. Find the area enclosed between the two parabolas $y = x^2 - 4x + 2$ and $y = 2 - x^2$.

Integration – Exponential Functions

1. Integrate the following:

(a) $\int 5e^{5x+3} dx$	(b) $\int 7e^{7x} + 3x^2 dx$	(c) $\int e^x + e^{-3x} dx$
(d) $\int e^{-x} dx$	(e) $\int e^{3x+4} dx$	(f) $\int \sqrt[3]{e^{5x}} dx$
(g) $\int e^3 dx$	(h) $\int \frac{x}{e^{x^2}} dx$	(i) $\int e^{-x} + e dx$
(j) $\int (e^x + e^{-x})^2 dx$	(k) $\int 5e^{-2.5x} dx$	(l) $\int \frac{e^{2x} + e^{5x}}{e^{2x}} dx$

2. Evaluate the following definite integrals:

(a) $\int_1^2 e^{2x-1} dx$	(b) $\int_{\ln 3}^{\ln 5} xe^{2x} dx$	(c) $5 \int_1^2 e^{3x} dx$
(d) $\int_0^2 -e^{-x} dx$	(e) $\int_0^{\ln 5} e^x dx$	(f) $\int_{-1}^3 e^{\frac{-x}{2}} dx$
(g) $\int_0^1 (e^{2x} - e^{-2x}) dx$	(h) $\int_{-3}^3 \left(e^{\frac{1}{2}} - e^{\frac{-x}{2}} \right) dx$	

Week 5C

Key Idea - Integration of Logarithmic Functions

1. Integrate the following:

(a) $\int \frac{1}{x} dx$	(b) $\int \frac{2}{2x-3} dx$	(c) $\int \frac{-9x^2}{2-3x^3} dx$
(d) $\int x + \frac{1}{x-3} dx$	(e) $\int \frac{3x^2}{x^3+5} dx$	(f) $\int \frac{3}{x} dx$
(g) $\int \frac{x}{x^2+2} dx$	(h) $\int \frac{x^2}{2x^3-7} dx$	(i) $\int \frac{x^3}{x^4+3} dx$
(j) $\int \frac{4x}{2x^2+1} dx$	(k) $\int \frac{x^2-1}{x} dx$	(l) $\int \left(x - \frac{1}{x^2} \right)^2 dx$

2. Evaluate the following definite integrals:

(a) $\int_0^1 \frac{6x^5}{x^6+1} dx$	(b) $\int_{-2}^0 \frac{1}{5+2x} dx$	(c) $\int_5^7 \frac{x-1}{x^2-2x} dx$
(d) $\int_{e^2}^e \frac{5}{x} dx$	(e) $\int_1^2 \frac{12}{12x+1} dx$	(f) $\int_2^4 \left(\frac{1}{x} + \frac{1}{x^2} \right) dx$

Integration – Trigonometric Functions

1. Integrate the following:

$$\begin{array}{lll} \text{(a)} \int 2 \sin(2x-1) dx & \text{(b)} \int 3 \sec^2(3x) - x dx & \text{(c)} \int \frac{1}{3} \cos\left(\frac{x}{3}\right) + 5 dx \\ \text{(d)} \int \sin(2x+5) dx & \text{(e)} \int \cos(3x) dx & \text{(f)} \int \sec^2(7x-1) dx \\ \text{(g)} \int 6 \sin\left(x + \frac{\pi}{4}\right) dx & \text{(h)} \int \frac{\cos(2x+1)}{2} dx & \text{(i)} \int 3 \sec^2\left(3 - \frac{\pi}{2}x\right) dx \end{array}$$

2. Evaluate the following definite integrals:

$$\begin{array}{lll} \text{(a)} \int_0^{\pi} \cos x dx & \text{(b)} \int_0^{\frac{\pi}{3}} \sec^2 x dx & \text{(c)} \int_0^{\pi} (\sin x + \cos x) dx \\ \text{(d)} \int_0^{\frac{\pi}{3}} 2 \sin(3x) dx & \text{(e)} \int_{-\pi}^0 3 \cos(x + \pi) dx & \text{(f)} \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} 4 \sec^2 x dx \\ \text{(g)} \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (\cos x - \sin 2x) dx & \text{(h)} \int_{\frac{\pi}{8}}^{\frac{\pi}{4}} (\sin 2x - \cos 2x) dx & \text{(i)} \int_0^{\frac{\pi}{3}} \left(3 \cos 3x - \frac{\sin 2x}{2}\right) dx \end{array}$$

Integration – Mixed

1. Integrate the following:

$$\begin{array}{lll} \text{(a)} \int e^x - \sin x + x + 2 dx & \text{(b)} \int \cos x - \sec^2 x + \frac{1}{x} dx & \text{(c)} \int \sqrt{x} - e^{2x-1} + \sin(2x) dx \\ \text{(d)} \int \sin(-\pi x + 2) + \frac{x^2}{x^3 + 1} dx & \text{(e)} \int 4 \cos(2x-1) + 7e^{5x-2} dx & \text{(f)} \int \left(\cos \frac{x}{2} - 2 \sin 2x\right) dx \\ \text{(g)} \int \left(\sqrt{x} - \frac{1}{2} \cos x\right) dx & \text{(h)} \int (x^2 + \sin 2x) dx \end{array}$$

Week 5D

Key Idea – Simple Substitution

1. Use substitution to integrate the following:

$$\begin{array}{lll} \text{(a)} \int 2x(x^2-1)^4 dx & \text{(b)} \int 3x^2(x^3+5)^4 dx & \text{(c)} \int x^6(x^7-3)^5 dx \\ \text{(d)} \int (3-2x)^6 dx & \text{(e)} \int \frac{2x+4}{(x^2+4x+7)^5} dx & \text{(f)} \int \frac{1}{(2x+3)^3} dx \\ \text{(g)} \int 2x^3 \sqrt{x^2+1} dx & \text{(h)} \int 2x \sqrt{1-x^2} dx & \text{(i)} \int \frac{3}{(3x-5)^{\frac{2}{3}}} dx & \text{(j)} \int \frac{x-1}{\sqrt{x^2-2x+3}} dx \end{array}$$

2. Using substitution to evaluate the following definite integrals:

$$\begin{array}{lll} \text{(a)} \int_0^1 5x^4(x^5-1)^3 dx & \text{(b)} \int_0^2 x^2(x^3+2)^3 dx & \text{(c)} \int_{-1}^1 (1+x)^{1/2} dx & \text{(d)} \int_0^2 \frac{1}{(3x-5)^{\frac{2}{3}}} dx \end{array}$$